

Data warehouses

Power Quality Data Mart: Integrating and Analysing E-REDES Operational Data

Master in Data Science and Engineering

Carolina Dias, up202500534

Leonor Couto, up202205796

Mariana Pereira, up202205093

Simão Bernardo, up202502529

Sofia Fernandes, up202502530

March 2026

Contents

1. Subject Description and Goals	4
2. Data Context	4
3. Planning	6
3.1. Dimensional Bus Matrix	6
3.2. Dimension Hierarchy Notes	6
3.3. Data Dictionary.....	7
4. Dimensional Data Model	7
5. Extraction, Transformation and Loading (ETL)	9
5.1. Data Sources Selection	9
5.2. Data Extraction	9
5.3. Data Transformation	9
5.4. Data Loading	10
6. Querying and Data Analysis	10
Q1: Exceptional Events vs. Power Quality Degradation (vw_exceptional)	11
Q2: Flicker Non-Compliance by Municipality and Voltage Level (vw_non_compliance).....	11
Q3: Severity-Weighted Voltage Event Index by NUTS II Region (vw_severity_index)	12
Q4: Longitudinal Power Quality Trend (vw_annual_trends).....	12
Q5: Installation Type Comparison: PTD vs. Substation (vw_installation_comparison).....	13
Q6: Phase Asymmetry Analysis (vw_phase_asymmetry)	13
Q7: Power Quality vs. Installation Load Size (vw_load_size_comparison).....	14

7. Visualization	14
8. Concluding Remarks.....	16
9. References	17
10.Appendix.....	18
10.1 Fact Tables.....	18
10.2 Dimension Tables.....	21

1. Subject Description and Goals

This project focuses on the design and implementation of a Data warehouse to analyse power quality phenomena in Portugal's electricity distribution network, using data provided by the E-REDES Open Data Portal [1].

The Data warehouse integrates multiple datasets from the E-REDES Open Data Portal, combining infrastructure information with power quality measurements to support comprehensive analysis.

The central analytical focus of the project is the study of voltage disturbances and compliance with the NP EN 50160 standard, which defines acceptable limits for power quality in electrical networks. By combining event-based and continuous monitoring data, the system enables a comprehensive evaluation of network performance across different locations, voltage levels, and time periods.

The main goal of this project is to design and implement a Data warehouse that supports efficient multidimensional analysis of power quality data, fulfilling the requirements of analytical querying and aggregation defined in the assignment.

Specifically, the project aims to:

- Integrate heterogeneous datasets from the E-REDES Open Data Portal into a unified analytical structure
- Design a dimensional model with multiple fact tables, including both transactional (event-level) and aggregated (snapshot) data
- Support the analysis of voltage events (dips and overvoltages) by duration, magnitude, location, and voltage level
- Enable the evaluation of continuous power quality indicators and their compliance with regulatory standards
- Provide aggregated views at municipality level, allowing efficient regional and temporal analysis
- Ensure the presence of additive and semi-additive measures, in accordance with assignment requirements
- Demonstrate how a data warehouse enables analytical capabilities that are not easily achievable with operational data sources

Overall, the project delivers a structured and scalable analytical system that supports both detailed and high-level analysis of power quality, enabling insights into the behaviour and reliability of the electricity distribution network.

2. Data Context

The Data warehouse is built using datasets from the E-REDES Open Data Portal, which provide complementary perspectives on Portugal's electricity distribution network.

In total, **five datasets** were selected as the primary data sources for this project:

- *Network Features*, containing technical characteristics of substations and network infrastructure;

- *Electrical Power Quality - Voltage Dips*, describing voltage dips;
- *Electrical Power Quality - Overvoltages*, describing overvoltage events;
- *Electrical Power Quality - Continuous Phenomena*, providing continuous power quality indicators;
- *Secondary Substations (PTD)*, containing information on distribution transformer stations.

The data can be broadly categorized into three main groups:

- **Infrastructure data**, describing electrical installations such as substations and transformer stations, including their technical characteristics, capacity, and geographic location;
- **Voltage event data**, capturing disturbances such as voltage dips and overvoltages, classified according to magnitude and duration based on the NP EN 50160 standard, using predefined categories (e.g., A1-X5 for voltage dips and S1-T3 for overvoltages) that represent different combinations of voltage variation and event duration);
- **Continuous power quality data**, providing indicators such as voltage compliance, flicker, harmonic distortion, and frequency stability, expressed as percentages of time within acceptable limits.

These datasets are combined to enable the analysis of power quality both from an event-based perspective and a continuous monitoring perspective, across different locations, voltage levels, and time periods.

To support spatial and temporal analysis, the data is enriched with administrative geographic information, such as information on municipalities and regional classifications, and a structured time dimension derived from the temporal attributes present in the source datasets.

It should be noted that the data provided by E-REDES represents approximations of network conditions at the time of collection and may be subject to minor inconsistencies. Nevertheless, it provides a solid basis for analytical modelling and exploration.

3. Planning

3.1. Dimensional Bus Matrix

The dimensional bus matrix is a planning and documentation tool used in Data warehouse design to map the relationship between the fact tables, or stars, and their associated dimensions. It provides a concise overview of which dimensions are shared across the different stars within a data mart, enabling the identification of conformed dimensions that ensure consistency and integration across analytical processes.

The following table (Table 1) presents the dimensional bus matrix for the **Technical Power Quality** data mart, which comprises four stars: *VoltageEvent*, *ContinuousPhenomena*, *Agg_Municipality_Voltage_Event*, and *Agg_Municipality_Continuous_Quality*. Each column represents a dimension, and each mark indicates that the respective star makes use of that dimension.

Table 1 - Bus Matrix

Data mart	Star	Dimension	Installation	Voltage Class	Voltage Level	Exceptional Event	Date	Location
Technical Power Quality	VoltageEvent		x	x	x	x	x	x
	ContinuousPhenomena		x		x	x	x	x
	Agg_Municipality_Voltage_Event			x	x		x	x
	Agg_Municipality_Continuous_Quality				x		x	x

The *Date* and *Location* dimensions are shared across all four stars, making them fully conformed dimensions within this data mart. *VoltageLevel* is also present in all four stars. *Installation* and *ExceptionalEvent* appears only in the two base-level stars, *VoltageEvent* and *ContinuousPhenomena*, as the aggregated stars operate at the municipality level and no longer carry installation-level or event-type granularity. *VoltageClass* is used in *VoltageEvent* and *Agg_Municipality_Voltage_Event*, reflecting its relevance specifically to voltage event analysis.

3.2. Dimension Hierarchy Notes

The bus matrix also defines the **structure of each dimension**, including hierarchical levels where applicable, which determines the granularity at which drill-down and roll-up operations can be performed.

- **Dim_Date:** *Full_Date* → *Day* → *Month* → *Quarter* → *Year*. This dimension supports temporal analysis at multiple levels of granularity, allowing aggregation from daily observations up to yearly totals.
- **Dim_Location:** This dimension captures the Portuguese administrative and statistical geography and supports **multiple hierarchical roll-up paths**. One hierarchy follows the administrative structure: *Municipality* → *District*. A second hierarchy follows the European statistical classification: *Municipality* → *NUTS III* → *NUTS II*. These alternative hierarchies allow spatial analysis to be performed according to different aggregation perspectives, depending on the analytical requirements.

The remaining dimensions do not present a hierarchical structure, but can be described as follows:

- **Dim_Installation:** A flat dimension with descriptive attributes such as *Installation_Code*, *Designation*, *Type_of_Installation*, *Busbar*, and *Usage_Level_Band*, among others. These attributes characterize the electrical installation but do not form a natural aggregation path.
- **Dim_Voltage_Class:** Contains attributes such as *Voltage_Event_Type*, *Class_Family_Code*, and *Duration_Class_Code*, which describe different characteristics of voltage events. These attributes provide multiple classification perspectives for analysis rather than a strict hierarchical structure.
- **Dim_Voltage_Level:** A flat dimension which includes attributes such as *Nominal_Voltage_V* and *Voltage_Band_Code*, used to group installations at voltage level. These attributes do not define a hierarchical structure.
- **Dim_Exceptional_Event:** A simple dimension consisting of *Exceptional_Event_ID* and *Exceptional_Event_Name*, used to identify external events that may influence power quality. It does not contain hierarchical levels.

3.3. Data Dictionary

The data dictionary provides a detailed description of all fact and dimension tables that compose the Technical Power Quality data mart. It includes attribute names, data types, key classifications, and hierarchical levels where applicable.

The data dictionary includes the four fact tables *Fact_Voltage_Event*, *Fact_Continuous_Phenomena*, *Agg_Municipality_Voltage_Event*, and *Agg_Municipality_Continuous_Quality*, as well as the six dimension tables *Dim_Date*, *Dim_Location*, *Dim_Installation*, *Dim_Voltage_Class*, *Dim_Voltage_Level*, and *Dim_Exceptional_Event*.

Due to the big volume of information, the complete data dictionary is provided in the appendix at the end of this report.

4. Dimensional Data Model

The dimensional data model, illustrated in Figure 1, implements the Technical Power Quality data mart defined in the planning phase. The model contains two transaction fact tables and two periodic snapshot aggregates.

5. Extraction, Transformation and Loading (ETL)

5.1. Data Sources Selection

The datasets were all selected from the E-Redes Open Data Portal and downloaded as CSV (comma-separated value) files, as the data on this platform is made publicly available and the datasets mentioned above are directly aligned with the project's analytical goals. The five datasets were identified as the basis for technical analysis.

Complementary to the operational data, we incorporated auxiliary administrative and spatial datasets, thus ensuring a robust and standardized geographic dimension. NUTS III regional definitions and codes were obtained from the European Commission's GISCO (Geographic Information System of the Commission) distribution service via a GeoJSON export, while standard municipality names and codes were sourced from the Eurostat Local Administrative Units (LAU) registry [2].

5.2. Data Extraction

The operational E-REDES datasets were manually exported from the Open Data portal as CSV files. Similarly, the Eurostat LAU registry was extracted in tabular format, while the spatial NUTS III definitions were extracted from a compressed GeoJSON file. These static files established a localized staging area, by providing a stable snapshot of historical data (2014-2025). The raw data was combined in a single excel file with multiple sheets for better understanding and usage. From this staging area, the data was ingested into processing scripts to undergo cleansing and dimensional restructuring.

5.3. Data Transformation

The transformation phase was implemented using Python, using the well-known Pandas library to clean, integrate, and reshape the raw data into the proposed dimensional schema. Preliminary pre-processing applied across all datasets included string standardization, the removal of trailing whitespaces, and the conversion of improper null values to true null values. Base data for the dimensions and facts was gathered by concatenating relevant columns from multiple sources, filtering out anomalous values, and resolving missing data prior to structural reshaping.

Creating the dimension tables involved some common transformation steps, such as joining datasets by concatenated columns and generating surrogate keys. However, due the heterogeneity of the data, distinct transformation strategies were required. Complex columns were parsed into granular attributes, such as the splitting of coordinates into latitude and longitude in Dim_Installation and the extraction of hierarchical temporal attributes from full date strings in Dim_Date. Custom functions and dictionaries were applied to categorize raw numerical voltages into ranks, in Dim_Voltage_Level, and to correct anomalous descriptions for exceptional events Dim_Exceptional_Event, while auxiliary datasets were utilized to cross-reference and impute missing regional geographic codes in Dim_Location. Conversely, the voltage class dimension was generated by mapping official EN 50160 and ERSE standard dictionaries to establish residual voltage ranges, duration classes, and severity risk levels, rather than extracting these directly from the raw data. To guarantee referential integrity within the Data warehouse, a default record was introduced into all dimensions to account for unknown parameters or standard conditions.

Populating the fact tables required mapping the raw data to the newly generated dimensional surrogate keys and implementing an ID default value of 0 for unmapped foreign keys to prevent data loss. When dealing with continuous data, the discrete foreign keys (e.g., voltage class key) were assigned a predefined value of -1, representing its inapplicability to continuous monitoring. These fact tables were reshaped depending on the nature of the observations. The discrete voltage event dataset was unpivoted, by transforming horizontal voltage class columns into distinct, vertical rows representing event counts. In contrast, the continuous phenomena dataset maintained its original tabular structure but required the calculation of new metrics. This included a monitoring days measure and a compliance metric created by averaging various power quality percentages variables.

To facilitate high-level temporal and regional reporting, periodic snapshot tables were ultimately aggregated from the base transaction facts. As a first step, the transaction tables were joined with the temporal dimension to retrieve monthly identifiers. The granular data was then grouped at a predefined intersection of month, municipality, voltage level and voltage class. Additionally, higher-level analytical measures were computed through grouping operations. These aggregations calculated the sum of event occurrences and accumulated monitoring days, the mathematical averages of the continuous power quality indicators, and the distinct count of unique active installations, thereby completing the regional profiles and finalizing the data types for analytical readiness.

5.4. Data Loading

The final phase of the ETL pipeline involved loading the dataframes into a target MySQL relational database. This process was automated using the Python SQLAlchemy library in conjunction with the PyMySQL database driver. To preserve referential integrity and comply with the constraints defined in the physical data model, a hierarchical loading sequence was enforced. Initially, all dimension tables were populated to establish the primary surrogate keys necessary for relational mapping. Before loading the factual data, a schema alignment procedure was executed: redundant attributes, such as denormalized geography and classification IDs temporarily held in the staging DataFrames, were dropped. This ensured the data conformed to the normalized physical Star Schema defined in the database architecture. Subsequently, the transactional fact tables recording voltage events and continuous phenomena were loaded, followed by the aggregated periodic snapshots. The physical MySQL schema enforces Foreign Key constraints across these relationships, thus, guaranteeing that no factual record is unlinked and that the Data warehouse maintains strict structural integrity for all downstream analytical querying and visualization tasks.

6. Querying and Data Analysis

To validate the dimensional model and demonstrate the analytical capabilities of the Data warehouse, a set of seven analytical queries was developed, each addressing a specific business question related to power quality monitoring in Portugal. These queries exploit the star schema's conformed dimensions to enable consistent aggregation across time, geography, and infrastructure.

The design of the queries reflects the distinction between fully additive and semi-additive measures. Voltage event counts can be aggregated directly across dimensions, while power quality indicators (e.g., flicker and harmonic distortion) require weighted averages based on monitoring duration to ensure statistical correctness. Together, these queries form a

coherent analytical layer that supports both detailed investigation and higher-level exploration, later consumed by visualization tools.

Q1: Exceptional Events vs. Power Quality Degradation (vw_exceptional)

This query isolates monitoring periods flagged as exceptional events, such as storms and depressions, and quantifies their impact on voltage event counts. By computing the total number of events, monitoring duration, and a derived events-per-day rate for each named event, the query allows direct comparison of event intensity. This is critical for distinguishing baseline grid behaviour from weather-driven degradation, a key requirement for fair regulatory compliance reporting under NP EN 50160.

```

1  create or replace view vw_exceptional as
2
3  SELECT
4      ee.exceptional_event_name,
5      COUNT(*) AS records,
6      SUM(fve.event_count) AS total_events,
7      AVG(fve.event_count) AS avg_events_per_record,
8      SUM(d_end.full_date - d_start.full_date) AS total_duration_days,
9      SUM(fve.event_count) / NULLIF(SUM(d_end.full_date - d_start.full_date), 0) AS events_per_day,
10     AVG(fve.event_count / NULLIF((d_end.full_date - d_start.full_date), 0)) AS avg_events_per_day_per_record
11 FROM fact_voltage_event fve
12 JOIN dim_exceptional_event ee ON fve.exceptional_event_id = ee.exceptional_event_id
13 JOIN dim_date d_start ON fve.start_date_id = d_start.date_id
14 JOIN dim_date d_end ON fve.end_date_id = d_end.date_id
15 WHERE d_end.full_date > d_start.full_date
16 GROUP BY ee.exceptional_event_name
17 ORDER BY events_per_day DESC;

```

Figure 2 - SQL view definition for vw_exceptional

Q2: Flicker Non-Compliance by Municipality and Voltage Level (vw_non_compliance)

Query number 2 identifies regions with poorer power quality performance by aggregating flicker compliance at municipality and regional levels. The use of the worst-phase value ensures a conservative assessment, while the minimum monitoring threshold guarantees statistical reliability of the results.

```

1  create view vw_non_compliance as
2
3  SELECT
4      dl.municipality,
5      dl.nuts_ii,
6      dvl.voltage_band_description,
7      AVG(amcp.avg_flicker_l1_pct) AS flicker_l1,
8      AVG(amcp.avg_flicker_l2_pct) AS flicker_l2,
9      AVG(amcp.avg_flicker_l3_pct) AS flicker_l3,
10     LEAST(
11         AVG(amcp.avg_flicker_l1_pct),
12         AVG(amcp.avg_flicker_l2_pct),
13         AVG(amcp.avg_flicker_l3_pct)
14     ) AS worst_phase_flicker,
15     SUM(amcp.total_monitoring_days) AS total_monitoring_days,
16     SUM(amcp.installation_count) AS total_installation_months
17 FROM agg_municipality_continuous_phenomena amcp
18 JOIN dim_location dl ON amcp.municipality_id = dl.municipality_id
19 JOIN dim_voltage_level dvl ON amcp.voltage_level_id = dvl.voltage_level_id
20 GROUP BY dl.municipality, dl.nuts_ii, dvl.voltage_band_description
21 HAVING SUM(amcp.total_monitoring_days) >= 365
22 ORDER BY worst_phase_flicker ASC;

```

Figure 3 - SQL view definition for vw_non_compliance

Q3: Severity-Weighted Voltage Event Index by NUTS II Region (vw_severity_index)

To move beyond simple event counting, this query introduces a severity-weighted index, assigning higher importance to more critical disturbances. Aggregated at the NUTS II level, this metric enables meaningful regional comparison and supports prioritisation of intervention areas.

```
1 create or replace view vw_severity_index as
2
3 SELECT
4     dl.nuts_ii,
5     SUM(amve.total_event_count) AS total_events,
6     SUM(amve.total_event_count * dvc.severity_rank) AS weighted_severity_index,
7     SUM(amve.total_event_count * dvc.severity_rank) / NULLIF(SUM(amve.total_event_count), 0) AS avg_severity_per_event,
8     COUNT(DISTINCT amve.municipality_id) AS municipalities_covered,
9     SUM(amve.total_record_count) AS total_records
10 FROM agg_municipality_voltage_event amve
11 JOIN dim_location dl ON amve.municipality_id = dl.municipality_id
12 JOIN dim_voltage_class dvc ON amve.voltage_class_id = dvc.voltage_class_id
13 GROUP BY dl.nuts_ii
14 ORDER BY weighted_severity_index DESC;
```

Figure 4 – SQL view definition for vw_severity_index

Q4: Longitudinal Power Quality Trend (vw_annual_trends)

This query analyses the temporal evolution of key power quality indicators, computing yearly trends based on weighted averages. By tracking worst-phase values over time, it provides a longitudinal perspective essential for identifying systemic improvements or degradation patterns.

```
1 create or replace view vw_annual_trends as
2 SELECT
3     dd.year,
4     AVG(amcp.avg_flicker_l1_pct) AS avg_flicker_l1,
5     AVG(amcp.avg_flicker_l2_pct) AS avg_flicker_l2,
6     AVG(amcp.avg_flicker_l3_pct) AS avg_flicker_l3,
7     LEAST(
8         AVG(amcp.avg_flicker_l1_pct),
9         AVG(amcp.avg_flicker_l2_pct),
10        AVG(amcp.avg_flicker_l3_pct)
11    ) AS worst_phase_flicker,
12
13     AVG(amcp.avg_thd_l1_pct) AS avg_thd_l1,
14     AVG(amcp.avg_thd_l2_pct) AS avg_thd_l2,
15     AVG(amcp.avg_thd_l3_pct) AS avg_thd_l3,
16     LEAST(
17         AVG(amcp.avg_thd_l1_pct),
18         AVG(amcp.avg_thd_l2_pct),
19         AVG(amcp.avg_thd_l3_pct)
20    ) AS worst_phase_thd,
21
22     -- Voltage Averages
23     AVG(amcp.avg_voltage_l1_pct) AS avg_voltage_l1,
24     AVG(amcp.avg_voltage_l2_pct) AS avg_voltage_l2,
25     AVG(amcp.avg_voltage_l3_pct) AS avg_voltage_l3,
26     LEAST(
27         AVG(amcp.avg_voltage_l1_pct),
28         AVG(amcp.avg_voltage_l2_pct),
29         AVG(amcp.avg_voltage_l3_pct)
30    ) AS worst_phase_voltage,
31
32     -- Aggregated Totals
33     SUM(amcp.total_monitoring_days) AS total_monitoring_days,
34     SUM(amcp.installation_count) AS total_installation_months
35
36 FROM agg_municipality_continuous_phenomena amcp
37 JOIN dim_date dd ON amcp.month_id = dd.month_id
38 GROUP BY dd.year
39 ORDER BY dd.year ASC;
```

Figure 5 – SQL view definition for vw_annual_trends

Q5: Installation Type Comparison: PTD vs. Substation (vw_installation_comparison)

This query compares power quality performance between the two installation types, PTDs (distribution transformer stations) and Substations, broken down by municipality and voltage level. Weighted averages for flicker and voltage compliance are computed per installation type, and voltage event counts and average severity are joined from the voltage event fact table. This enables direct assessment of whether secondary low-voltage infrastructure systematically underperforms relative to primary medium-voltage substations.

```
1 create view vw_installation_comparison as
2 SELECT
3     dl.municipality, -- Added Geographic context
4     di.type_of_installation,
5     dvl.voltage_band_description,
6     SUM(fcp.flicker_l1_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_flicker_l1,
7     SUM(fcp.flicker_l2_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_flicker_l2,
8     SUM(fcp.flicker_l3_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_flicker_l3,
9     SUM(fcp.voltage_l1_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_voltage_l1,
10    SUM(fcp.voltage_l2_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_voltage_l2,
11    SUM(fcp.voltage_l3_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_voltage_l3,
12    COUNT(*) AS records
13 FROM fact_continuous_phenomena fcp
14 JOIN dim_installation di ON fcp.installation_id = di.installation_id
15 JOIN dim_voltage_level dvl ON fcp.voltage_level_id = dvl.voltage_level_id
16 JOIN dim_location dl ON fcp.municipality_id = dl.municipality_id -- Added JOIN for municipality
17 WHERE fcp.flicker_l1_pct < 100
18    AND fcp.flicker_l2_pct < 100
19    AND fcp.flicker_l3_pct < 100
20 GROUP BY
21     dl.municipality,
22     di.type_of_installation,
23     dvl.voltage_band_description
24 HAVING COUNT(*) >= 20
25 ORDER BY
26     dl.municipality,
27     di.type_of_installation,
28     dvl.voltage_band_description;
```

Figure 6 - SQL view definition for vw_installation_comparison

Q6: Phase Asymmetry Analysis (vw_phase_asymmetry)

Electrical installations ideally exhibit balanced performance across all three phases. This query detects asymmetry by computing weighted flicker and voltage compliance separately for L1, L2, and L3, grouped by installation type and voltage band. Records where all three phases simultaneously reach 100% compliance are excluded to focus the analysis on installations exhibiting measurable variation, with a minimum record threshold of 20 applied for statistical robustness.

```
1 create view vw_phase_asymmetry as
2
3 SELECT
4     di.type_of_installation,
5     dvl.voltage_band_description,
6     SUM(fcp.flicker_l1_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_flicker_l1,
7     SUM(fcp.flicker_l2_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_flicker_l2,
8     SUM(fcp.flicker_l3_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_flicker_l3,
9     SUM(fcp.voltage_l1_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_voltage_l1,
10    SUM(fcp.voltage_l2_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_voltage_l2,
11    SUM(fcp.voltage_l3_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_voltage_l3,
12    COUNT(*) AS records
13 FROM fact_continuous_phenomena fcp
14 JOIN dim_installation di ON fcp.installation_id = di.installation_id
15 JOIN dim_voltage_level dvl ON fcp.voltage_level_id = dvl.voltage_level_id
16 WHERE fcp.flicker_l1_pct < 100
17    AND fcp.flicker_l2_pct < 100
18    AND fcp.flicker_l3_pct < 100
19 GROUP BY di.type_of_installation, dvl.voltage_band_description
20 HAVING COUNT(*) >= 20
21 ORDER BY di.type_of_installation, dvl.voltage_band_description;
```

Figure 7 – SQL view definition for vw_phase_asymmetry

Q7: Power Quality vs. Installation Load Size (vw_load_size_comparison)

This query investigates whether installed transformer capacity, expressed in kVA, correlates with power quality outcomes. Installations are grouped into four power bands (Small, Medium, Large, and Very Large), and weighted flicker compliance and voltage unbalance are computed per band alongside the average number of clients served. This analysis supports infrastructure planning decisions by revealing whether larger or more heavily loaded installations are associated with higher rates of non-compliance.

```
1  create or replace view vw_load_size_comparison as
2
3  SELECT
4    CASE
5      WHEN di.installed_power_kva < 400 THEN '1 - Small (<400 kVA)'
6      WHEN di.installed_power_kva < 1260 THEN '2 - Medium (400-1260 kVA)'
7      WHEN di.installed_power_kva < 5000 THEN '3 - Large (1260-5000 kVA)'
8      ELSE '4 - Very Large (>5000 kVA)'
9    END AS power_band,
10   COUNT(DISTINCT fcp.installation_id) AS installations,
11   SUM(fcp.flicker_l1_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_avg_flicker,
12   SUM(fcp.unbalanced_pct * fcp.monitoring_days) / NULLIF(SUM(fcp.monitoring_days), 0) AS weighted_avg_unbalance,
13   AVG(di.number_of_clients) AS avg_clients_served,
14   SUM(fcp.monitoring_days) AS total_monitoring_days
15 FROM fact_continuous_phenomena fcp
16 JOIN dim_installation di ON fcp.installation_id = di.installation_id
17 WHERE di.installed_power_kva IS NOT NULL
18    AND fcp.flicker_l1_pct < 100
19 GROUP BY
20   CASE
21     WHEN di.installed_power_kva < 400 THEN '1 - Small (<400 kVA)'
22     WHEN di.installed_power_kva < 1260 THEN '2 - Medium (400-1260 kVA)'
23     WHEN di.installed_power_kva < 5000 THEN '3 - Large (1260-5000 kVA)'
24     ELSE '4 - Very Large (>5000 kVA)'
25   END
26 ORDER BY power_band;
```

Figure 8 - SQL view definition for vw_load_size_comparison

7. Visualization

The visualization layer was implemented using Microsoft Power BI, illustrated in Figure 2, providing an interactive interface that builds directly on the analytical views defined in the querying layer. Rather than querying raw fact tables, the dashboard uses these precomputed views, ensuring consistency between the dimensional model and the visual analysis, while maintaining efficient performance.

The dashboard enables exploration of power quality across Portugal between 2013 and 2025, combining high-level indicators with drill-down capabilities. Key performance indicators include total voltage events (both overall and during exceptional events), the number of monitored installations (PTDs and substations), and the number of municipalities covered. These indicators provide an immediate overview of network scale, infrastructure coverage, and monitoring scope.

A central feature of the dashboard is its hierarchical geographic analysis. Visualizations support drill-down from NUTS II to NUTS III and ultimately to municipality level, allowing users to transition from regional summaries to localized insights. This hierarchical navigation is complemented by a temporal analysis, where yearly trends reveal the evolution of power quality indicators and enable the identification of long-term patterns.

The analytical depth of the dashboard is further explored through a set of coordinated visualizations, each reflecting a specific analytical perspective derived from the underlying queries:

- **Power Quality by Region:** A grouped bar chart presents these same worst-phase indicators across NUTS II regions, supporting drill-down from NUTS II through NUTS

III to municipality level. Upon updating the level, both the *Power Quality Trend Over Time* and *Top 10 Municipalities by Voltage Events* charts update accordingly, enabling progressively granular analysis. Centro, Norte, and Oeste e Vale do Tejo stand out for their comparatively lower compliance levels, indicating potential areas of concern.

- **Power Quality Trend Over Time:** A yearly trend line chart that tracks the worst-phase values of key power quality indicators, namely flicker, total harmonic distortion (THD), and voltage compliance, across the three phases (L1, L2, L3). This visualization reveals a noticeable degradation around 2016, followed by a stabilization trend approaching full compliance in subsequent years.
- **Top 10 Municipalities by Voltage Events:** A ranked bar chart highlights the top 10 municipalities by total number of voltage events, with Ponte de Sor recording the highest count, supporting targeted local analysis.
- **Impact of Exceptional Events:** A treemap illustrates the impact of exceptional events, showing the relative contribution of each event to overall voltage disturbances. Notably, *Ciclone Kirk* emerges as the event with the highest impact, highlighting the influence of extreme weather conditions on network performance.
- **Severity of Voltage Events by Region:** A horizontal bar chart displays the severity-weighted voltage event index by NUTS II region, clearly identifying Alentejo and Norte as the regions with the highest disturbance intensity.

Together, these visualizations show how the dimensional model supports both aggregated monitoring and fine-grained analytical exploration. By structuring the Data Warehouse around conformed dimensions and analytically meaningful measures, the system allows a seamless transition from structured data to actionable insights.

Power Quality Monitoring in Portugal

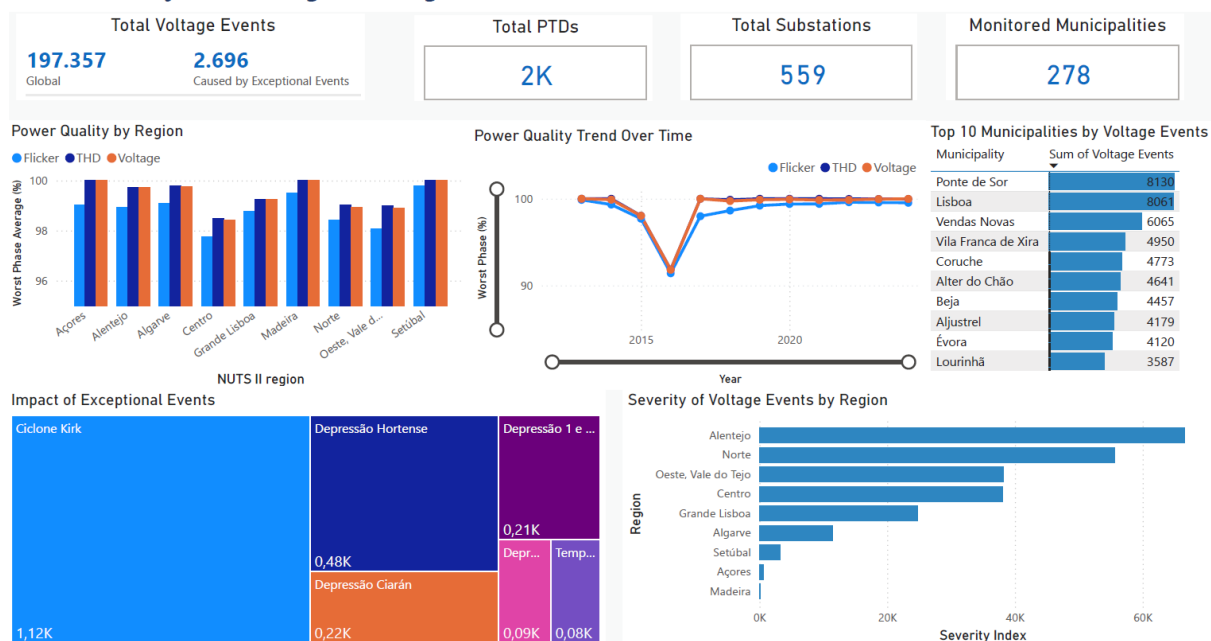


Figure 9 – Final Visualization Dashboard

8. Concluding Remarks

This project designed and implemented a Data warehouse system that integrates distinct datasets and facilitates complex and multidimensional analysis. By integrating a set of E-redes diagnostic and descriptive data, we were able to combine voltage event data, infrastructure and network characteristics into a unified system.

By combining transactional and aggregated data, our database supports multidimensional at various levels of detail (e.g., national and municipality level). The ETL process was performed after a thorough examination of the datasets, thus ensuring data quality and usability. Subsequently, the efficiency of this step was validated by running the analytical queries, as inconsistent results would evidence any referential mismatch.

As a result, the queries and visualizations created in Power BI provided patterns in power quality, including regional differences, temporal trends, and the impact of exceptional events.

All in all, our solution provided a structured basis for power quality analysis, thus emphasizing the benefit of data warehousing in extracting meaningful results from multidimensional data.

Future work could include applying machine learning models to the data we are using, to allow the system to learn to predict power outages, voltage dips or equipment failures before they happen.

9. References

- [1] E-REDES – Distribuição de Eletricidade, “E-REDES Open Data Portal.” [Online]. Available: <https://e-redes.opendatasoft.com/explore/?exclude.keyword=internal&sort=modified>. [Accessed: Mar. 23, 2026].
- [2] *Local Administrative Units (LAU)*. Local administrative units (LAU) - NUTS - Nomenclature of territorial units for statistics - Eurostat. (n.d.). <https://ec.europa.eu/eurostat/web/nuts/local-administrative-units> [Accessed: Mar. 23, 2026].

10. Appendix

This appendix presents the data dictionary of the dimensional model developed for power quality monitoring. It covers all fact and dimension tables and describes for each one its grain, attribute, data types and keys.

10.1 Fact Tables

Fact tables record the core measurement events captured per installation and measurement period. Aggregate tables summarize that same data at the municipality level to support broader analytical queries. Each table is described by its grain, meaning what a single row represents, followed by its dimensions and measures.

Star	Fact_Voltage_Event	Versio n	1.0	Date	02-04-2026
Granularity	One row per installation × measurement period × voltage event				
Dimensions					
Installation	Dim_Installation				
Voltage Class	Dim_Voltage_Class				
Voltage Level	Dim_Voltage_Level				
Exceptional Event	Dim_Exceptional_Event				
Start Date	Dim_Date				
End Date	Dim_Date				
Location	Dim_Location				
Measures					
Event_Count	Total number of voltage events recorded in the period				
Exceptional_Flag	1 if the period includes an exceptional event				
Record_Count	Total number of records				

Star	Fact_Continuous_Phenomena	Versio n	1.0	Date	02-04-2026
Granularity	One row per installation × measurement period × continuous phenomena assessment				
Dimensions					
Installation	Dim_Installation				
Voltage Level	Dim_Voltage_Level				
Exceptional Event	Dim_Exceptional_Event				
Start Date	Dim_Date				

End Date	Dim_Date				
Location	Dim_Location				
Measures					
Voltage_L1_Pct	Percentage of time voltage L1 within limits				
Voltage_L2_Pct	Percentage of time voltage L2 within limits				
Voltage_L3_Pct	Percentage of time voltage L3 within limits				
Flicker_L1_Pct	Percentage of time flicker L1 within limits				
Flicker_L2_Pct	Percentage of time flicker L2 within limits				
Flicker_L3_Pct	Percentage of time flicker L3 within limits				
THD_L1_Pct	Percentage of time THD L1 within limits				
THD_L2_Pct	Percentage of time THD L2 within limits				
THD_L3_Pct	Percentage of time THD L3 within limits				
Unbalanced_Pct	Percentage of time voltage unbalance within limits				
Frequency_Pct	Percentage of time frequency within limits				
Monitoring_Days	Number of days monitored in the period				
Norm_Compliance	Overall compliance indicator with EN 50160				

Star	Agg_Municipality_Voltage_Event	Version	1.0	Date	02-04-2026
Granularity	One row per municipality × month × voltage level × voltage class				
Dimensions					
Location	Dim_Location				
Voltage Level	Dim_Voltage_Level				
Voltage Class	Dim_Voltage_Class				
Month	Dim_Date				
Measures					
Total_Event_Count	Total number of voltage events aggregated at municipality level				
Total_Record_Count	Total number of records aggregated at municipality level				
Installation_Count	Number of installations contributing to the aggregate				

Star	Agg_Municipality_Cont_Quality	Version	1.0	Date	02-04-2026
Granularity	One row per municipality × month × voltage level				
Dimensions					
Location	Dim_Location				
Voltage Level	Dim_Voltage_Level				
Month	Dim_Date				
Measures					
Avg_Voltage_L1_Pct	Average % of time voltage L1 within limits across installations				
Avg_Voltage_L2_Pct	Average % of time voltage L2 within limits across installations				
Avg_Voltage_L3_Pct	Average % of time voltage L3 within limits across installations				
Avg_Flicker_L1_Pct	Average % of time flicker L1 within limits across installations				
Avg_Flicker_L2_Pct	Average % of time flicker L2 within limits across installations				
Avg_Flicker_L3_Pct	Average % of time flicker L3 within limits across installations				
Avg_THD_L1_Pct	Average % of time THD L1 within limits across installations				
Avg_THD_L2_Pct	Average % of time THD L2 within limits across installations				
Avg_THD_L3_Pct	Average % of time THD L3 within limits across installations				
Avg_Unbalanced_Pct	Average % of time voltage unbalance within limits across installations				
Avg_Frequency_Pct	Average % of time frequency within limits across installations				
Total_Monitoring_Days	Total monitoring days aggregated at municipality level				
Installation_Count	Number of installations contributing to the aggregate				

10.2 Dimension Tables

Dimension tables provide the descriptive context associated with each fact. They describe the installations, locations, dates, voltage classifications, and exceptional events that qualify the measurements recorded in the fact and aggregate tables.

Name	Description	SCD	Version	1.0	Date	02-04-2026
Dim_Date		Type 1	Hierarchy	Year > Quarter > Month > Day > Full_Date		
Attribute	Description	Level	Key	Type	Size	Precision
date_id	Surrogate key	Date	PK	INT		
full_date	Full calendar date	Date	UK	DATE		
day	Day of the month	Day		INT		
month_id	Foreign key to month level	Month	LK	INT		
month	Month number	Month	UK	INT		
month_name	Month name	Month		VARCHAR	20	
quarter_id	Foreign key to quarter level	Quarter	LK	INT		
quarter	Quarter number	Quarter	UK	INT		
year_id	Foreign key to year level	Year	LK	INT		
year	Calendar year	Year	UK	INT		
week_number	Week number of the year (derived attribute)	Date		INT		
day_of_week	Day of the week (derived attribute)	Date		VARCHAR	20	
is_weekend	Flag indicating whether the date is a weekend	Date		BOOLEAN		

Name	Description	SCD	Version	1.0	Date	02-04-2026
Dim_Location		Type 1	Hierarchy	Municipality > District / Municipality > NUTS III > NUTS II		
Attribute	Description	Level	Key	Type	Size	Precision
municipality_id	Surrogate key	Municipality	PK	INT		

municipality_code	Natural key of the municipality	Municipality	UK	VARCHAR	50	
municipality	Municipality name	Municipality		VARCHAR	255	
district_id	Foreign key to district level	District	LK	INT		
district_code	Natural key of the district	District	UK	VARCHAR	50	
district	District name	District		VARCHAR	255	
nuts_iii_id	Foreign key to NUTS III level	NUTS III	LK	INT		
nuts_iii_code	Natural key of the NUTS III region	NUTS III	UK	VARCHAR	50	
nuts_iii	NUTS III region name	NUTS III		VARCHAR	255	
nuts_ii_id	Foreign key to NUTS II level	NUTS II	LK	INT		
nuts_ii_code	Natural key of the NUTS II region	NUTS II	UK	VARCHAR	50	
nuts_ii	NUTS II region name	NUTS II		VARCHAR	255	

Name	Description	SCD	Version	1.0	Date	02-04-2026
Dim_Installation		Type 1	Hierarchy			
Attribute	Description	Level	Key	Type	Size	Precision
installation_id	Surrogate key	Installation	PK	INT		
installation_code	Natural key of the installation	Installation	UK	VARCHAR	50	
designation	Installation designation	Installation		VARCHAR	100	
type_of_installation	Type of electrical installation	Installation		VARCHAR	100	
busbar	Busbar identifier	Installation		VARCHAR	50	

construction_type	Construction type of the installation	Installation		VARCHAR	100	
usage_level_band	Usage level band	Installation		VARCHAR	50	
installed_power_kva	Installed power capacity in kVA	Installation		FLOAT		
contracted_power_kva	Contracted power in kVA	Installation		FLOAT		
number_of_clients	Number of clients served	Installation		INT		
power_generation_kw	Power generation capacity in kW	Installation		FLOAT		
number_of_client_producers	Number of client producers	Installation		INT		
latitude	Geographic latitude	Installation		FLOAT		
longitude	Geographic longitude	Installation		FLOAT		
at_mt_transformation_ratio_kv	Transformation ratio in kV	Installation		VARCHAR	50	
tip_mv	Tip power in MV	Installation		FLOAT		
max_sc_power_hv_mv	Maximum short-circuit power at HV in MV	Installation		FLOAT		
max_sc_power_mv_mv	Maximum short-circuit power at MV in MV	Installation		FLOAT		
min_sc_power_hv_mv	Minimum short-circuit power at HV in MV	Installation		FLOAT		
min_sc_power_mv_mv	Minimum short-circuit power at MV in MV	Installation		FLOAT		
sc_current_hv_ka	Short-circuit current at HV in kA	Installation		FLOAT		

Name	Description	SCD	Version	1.0	Date	02-04-2026
-------------	--------------------	------------	----------------	-----	-------------	------------

Dim_Voltage_Class		Type 1	Hierarchy			
Attribute	Description	Level	Key	Type	Size	Precision
voltage_class_id	Surrogate key	Voltage Class	PK	INT		
voltage_class_code	Natural key of the voltage class	Voltage Class	UK	VARCHAR	10	
voltage_event_type	Type of voltage event	Voltage Class		VARCHAR	50	
magnitude_basis	Basis for magnitude classification	Voltage Class		VARCHAR	50	
class_family_code	Family grouping of the voltage class	Voltage Class		VARCHAR	10	
duration_class_code	Duration classification code	Voltage Class		VARCHAR	10	
voltage_range_description	Description of the voltage range	Voltage Class		VARCHAR	100	
duration_range_description	Description of the duration range	Voltage Class		VARCHAR	100	
voltage_min_pct	Minimum voltage percentage threshold	Voltage Class		FLOAT		
voltage_max_pct	Maximum voltage percentage threshold	Voltage Class		FLOAT		
duration_min_ms	Minimum duration in milliseconds	Voltage Class		INT		
duration_max_ms	Maximum duration in milliseconds	Voltage Class		INT		
severity_rank	Severity ranking of the voltage class	Voltage Class		INT		

Name	Description	SCD	Version	1.0	Date	02-04-2026
Dim_Voltage_Level		Type 1	Hierarchy			
Attribute	Description	Level	Key	Type	Size	Precision
voltage_level_id	Surrogate key	Voltage Level	PK	INT		

nominal_voltage_v	Nominal voltage in volts	Voltage Level		FLOAT		
voltage_band_code	Code identifying the voltage band	Voltage Level		VARCHAR	20	
voltage_band_description	Description of the voltage band	Voltage Level		VARCHAR	100	

Name	Description	SCD	Version	1.0	Date	02-04-2026
Dim_Exceptional_Event		Type 1	Hierarchy			
Attribute	Description	Level	Key	Type	Size	Precision
exceptional_event_id	Surrogate key	Exceptional Event	PK	INT		
exceptional_event_name	Name or description of the exceptional event	Exceptional Event		VARCHAR	255	